ITIS4260 Project 1

January 26, 2025

1 Project 1: Domain Frequency Analysis

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[2]:
```python
# import pandas and sys libraries
import pandas as pd
import sys
```

[4]:
```python
# insert the path to the 'freq-master' directory into the system path
sys.path.insert(1, "./freq-master")

# import the FreqCounter class from the freq module
from freq import FreqCounter

# create an instance of the FreqCounter class and load the frequency table from
↪the file
freq_counter = FreqCounter()
freq_counter.load('./freq-master/freqtable2018.freq')
```

[6]:
```python
# read the csv file into a dataframe and display a few rows
df = pd.read_csv('top-1m.csv', header=None, index_col=False)
df.head()
```

[6]:

	0	1
0	1	google.com
1	2	microsoft.com
2	3	www.google.com
3	4	facebook.com
4	5	doubleclick.net

[8]:
```python
def domainNames(dataframe):
    # we created a list of all the values in column 1 of our dataframe which
↪are the URLs
    urlList = list(dataframe[1])
    # empty list to save the domain names
    domain_names = []
    for url in urlList:
        # creates a list with all the splits
```

```
        dsplit = url.split(".")
        # adding the domain name to the domain_names list above
        if len(dsplit) > 1:
            domain_names.append(dsplit[-2])
    return domain_names

# call the function
domains_list = domainNames(df)
```

[10]:

```
def frequency(domain_names):
    # domain names will be repeated many times so to calculate the frequency of␣
 ↪a domain name
    # we eliminate the duplicates to keep it consistant so we use set function␣
 ↪to do the same
    dnames_set= set(domain_names)
    # empty dictionary to save the frequency of the values in the above set
    set_dnames_freq= {}

    for dname in dnames_set:

        # the probability() function we used below is a function from the␣
 ↪frequencyCounter class we imported at the top
        # this probability function will return a list of average probability␣
 ↪and total word probability for a domain
        # we are interested in the average probability so we select the first␣
 ↪value from the list

        # adding calculated probability to the dictionary above

        set_dnames_freq[dname] = freq_counter.probability(dname)[0]

    # now lets get the above calculated freqency for all the domain names and␣
 ↪create a list
    # this will help us to map the freq with all the URLs in the dataset

    final_freq = []
    for dname in domain_names:
        # adding the frequency score to the list above
        final_freq.append(set_dnames_freq[dname])

    return final_freq

# call the function
totalFreq = frequency(domains_list)
```

[11]:
```python
# create a new dataframe with the URL, extracted domain name and the calculated␣
↪frequency for that domain
# below we used zip() function to map all the 3 lists.
frequency_df = pd.DataFrame(list(zip(list(df[1]),domains_list,totalFreq)),␣
↪columns = ['URL','Domain Name','Frequency'])

# print the first few rows of the frequency dataframe
print(frequency_df.head())
```

```
              URL  Domain Name  Frequency
0      google.com       google     6.6142
1   microsoft.com    microsoft     6.1558
2  www.google.com       google     6.6142
3    facebook.com     facebook     6.4581
4  doubleclick.net  doubleclick     7.7576
```

[12]:
```python
# filter the dataframe to count only frequencies under 3.5
count = frequency_df[frequency_df['Frequency'] < 3.5].shape[0]

# create a dataframe with domain names and their corresponding frequencies
df = pd.DataFrame({'Domain': domains_list, 'Frequency': totalFreq})

# filter the dataframe to include only frequencies under 3.5
filtered_df = df[df['Frequency'] < 3.5]

# print filtered dataframe and number of domains with frequencies less than 3.5
print(filtered_df)
print(f'Number of domains with frequency less than 3.5: {count}')
```

```
          Domain  Frequency
12         fbcdn     0.1321
20         fbcdn     0.1321
27        akadns     2.3848
34        akadns     2.3848
37         ytimg     2.9440
…            …          …
998590    shophq     3.4952
998628       smm     1.2489
998634  society6     2.6602
998637      sodu     2.8250
998677        ac     3.3945

[148824 rows x 2 columns]
Number of domains with frequency less than 3.5: 148824
```